

BC Geological Survey Assessment Report 42617



Ministry of Energy and Mines
BC Geological Survey

Assessment Report
Title Page and Summary

TYPE OF REPORT [type of survey(s)]: Geochemical

TOTAL COST: 11,633.45

AUTHOR(S): Matthew A. Temple

SIGNATURE(S): 

NOTICE OF WORK PERMIT NUMBER(S)/DATE(S): N/A

YEAR OF WORK: 2024

STATEMENT OF WORK - CASH PAYMENTS EVENT NUMBER(S)/DATE(S): 6046818 (December 6/2024) / 6058290 (Feb 19/2025)

PROPERTY NAME: New Sunro Copper Project

CLAIM NAME(S) (on which the work was done): 564267, 663263, 841971, 953676, 1016282, 1117931

COMMODITIES SOUGHT: Ag,Au,Cu,Pb,Zn,Mo

MINERAL INVENTORY MINFILE NUMBER(S), IF KNOWN: 092C 073, 092C 257, 092C 258, 092C 259, 092C 260

MINING DIVISION: Victoria Mining Division

NTS/BCGS: 092C 08E / 092C 050

LATITUDE: 48 ° 26 '49 " LONGITUDE: 124 ° 02 '29 " (at centre of work)

OWNER(S):

1) New Sunro Copper Ltd.

2)

MAILING ADDRESS:

55 Colorado Drive Campbell River, B.C. V9H 1N1

OPERATOR(S) [who paid for the work]:

1) New Sunro Copper Ltd.

2)

MAILING ADDRESS:

55 Colorado Drive Campbell River, B.C. V9H 1N1

PROPERTY GEOLOGY KEYWORDS (lithology, age, stratigraphy, structure, alteration, mineralization, size and attitude):

gabbro, diorite, diabase, basalt, agglomerate, flow breccia, tuff, Paleocene, Eocene, Oligocene, Metchosin Formation, Sooke Gabbro, Sheeted Dykes, High Level Gabbro, Carmanah Group, intrusions, dykes, folding, hornblende, actinolite, chlorite, urallite, epidote, zoisite, calcite, quartz, feldspar, magnetite, pyrite, pyrrhotite, chalcopryrite, native copper, copper, nickel, cobalt, silver, gold, molybdenite, shears, veinlets, smears, blebs, lenses, steeply dipping.

REFERENCES TO PREVIOUS ASSESSMENT WORK AND ASSESSMENT REPORT NUMBERS: 6592,211,27280,35982,40462,41634,27472, 30545,31362,31997,33480,2928,544,33931,4104.

TYPE OF WORK IN THIS REPORT	EXTENT OF WORK (IN METRIC UNITS)	ON WHICH CLAIMS	PROJECT COSTS APPORTIONED (incl. support)
GEOLOGICAL (scale, area)			
Ground, mapping _____			
Photo interpretation _____			
GEOPHYSICAL (line-kilometres)			
Ground			
Magnetic _____			
Electromagnetic _____			
Induced Polarization _____			
Radiometric _____			
Seismic _____			
Other _____			
Airborne _____			
GEOCHEMICAL (number of samples analysed for...)			
Soil _____			
Silt _____			
Rock 20 (plus 2 QAQC samples)			11,633,45
Other _____			
DRILLING (total metres; number of holes, size)			
Core _____			
Non-core _____			
RELATED TECHNICAL			
Sampling/assaying _____			
Petrographic _____			
Mineralographic _____			
Metallurgic _____			
PROSPECTING (scale, area) _____			
PREPARATORY / PHYSICAL			
Line/grid (kilometres) _____			
Topographic/Photogrammetric (scale, area) _____			
Legal surveys (scale, area) _____			
Road, local access (kilometres)/trail _____			
Trench (metres) _____			
Underground dev. (metres) _____			
Other _____			
		TOTAL COST:	11,633,45

Mineral Titles Online

Mineral Claim Exploration and Development Work/Expiry Date Change

Confirmation

Recorder: VIS, DARCY BEREND (280031)
Recorded: 2024/DEC/06
D/E Date: 2024/DEC/06

Submitter: VIS, DARCY BEREND (280031)
Effective: 2024/DEC/06

Confirmation

If you have not yet submitted your report for this work program, your technical work report is due in 90 days. The Exploration and Development Work/Expiry Date Change event number is required with your report submission. **Please attach a copy of this confirmation page to your report.** Contact Mineral Titles Branch for more information.

Event Number: 6046818

Work Type: Technical Work
Technical Items: Geochemical

Work Start Date: 2024/DEC/04
Work Stop Date: 2024/DEC/05
Total Value of Work: \$ 5212.25
Mine Permit No:

Summary of the work value:

Title Number	Claim Name	Issue Date	Good To Date	New Good To Date	# of Days Forward	Area in Ha	Applied Work Value
564267	SUNRO 1	2007/AUG/08	2025/JAN/22	2025/SEP/01	222	21.42	\$ 260.60
663263		2009/NOV/01	2025/JAN/22	2025/SEP/01	222	107.12	\$ 1303.01
841971		2010/DEC/30	2025/JAN/22	2025/SEP/01	222	128.54	\$ 1563.62
953676		2012/MAR/01	2025/JAN/22	2025/SEP/01	222	42.85	\$ 521.29
1016282		2013/JAN/24	2025/JAN/22	2025/SEP/01	222	42.85	\$ 521.28
1117931		2013/MAR/17	2025/JAN/22	2025/SEP/01	222	85.70	\$ 1042.45

Financial Summary:

Total applied work value: 5212.25

PAC name:

Note: Any PAC debit and credit amounts will be calculated after the assessment report has been submitted and approved.

Please print this page for your records.

The event was successfully saved.

Click [here](#) to return to the Main Menu.

Mineral Titles Online

Mineral Claim Exploration and Development Work/Expiry Date Change

Confirmation

Recorder: VIS, DARCY BEREND (280031)
Recorded: 2025/FEB/19
D/E Date: 2025/FEB/19

Submitter: VIS, DARCY BEREND (280031)
Effective: 2025/FEB/19

Confirmation

If you have not yet submitted your report for this work program, your technical work report is due in 90 days. The Exploration and Development Work/Expiry Date Change event number is required with your report submission. **Please attach a copy of this confirmation page to your report.** Contact Mineral Titles Branch for more information.

Event Number: 6058290

Work Type: Technical Work
Technical Items: Geochemical

Work Start Date: 2024/DEC/04
Work Stop Date: 2024/DEC/05
Total Value of Work: \$ 6421.20
Mine Permit No:

Summary of the work value:

Title Number	Claim Name	Issue Date	Good To Date	New Good To Date	# of Days Forward	Area in Ha	Applied Work Value
564267	SUNRO 1	2007/AUG/08	2025/SEP/01	2026/JUN/01	273	21.42	\$ 320.47
663263		2009/NOV/01	2025/SEP/01	2026/JUN/01	273	107.12	\$ 1602.36
841971		2010/DEC/30	2025/SEP/01	2026/JUN/01	273	128.54	\$ 1922.83
953676		2012/MAR/01	2025/SEP/01	2026/JUN/01	273	42.85	\$ 641.05
1016282		2013/JAN/24	2025/SEP/01	2026/JUN/01	273	42.85	\$ 641.05
1117931		2013/MAR/17	2025/SEP/01	2026/JUN/01	273	85.70	\$ 1281.94

Financial Summary:

Total applied work value: 6409.70

PAC name: New Sunro

Note: Any PAC debit and credit amounts will be calculated after the assessment report has been submitted and approved.

Please print this page for your records.

The event was successfully saved.

Click [here](#) to return to the Main Menu.

ASSESSMENT REPORT

on the

NEW SUNRO COPPER PROPERTY 2024

Event Number(s): 6046818, 6058290

Claims Worked On: 564267, 663263, 841971, 953676, 1016282, 1117931

Located in the
Victoria Mining Division
British Columbia, Canada

NTS Map Sheet: 092C 08E
BCGS Map Sheet: 092C 050

48° 26' 49" North Latitude | 124° 02' 29" West Longitude

Prepared for:

New Sunro Copper Ltd.
55 Colorado Drive
Campbell River, B.C. V9H 1N1

Prepared by:

Matthew Temple GIT

TRIPOINT GEOLOGICAL SERVICES LTD.

EGBC Permit to Practice: 1001503
148-2770 Leigh Road
Victoria BC, V9B 4G1

February 19, 2025

TABLE OF CONTENTS

1	INTRODUCTION.....	1
2	LOCATION, ACCESS, PHYSIOGRAPHY & CLIMATE.....	3
3	LAND TENURE AND CLAIM STATUS.....	4
4	EXPLORATION HISTORY	6
5	GEOLOGY.....	9
5.1	REGIONAL GEOLOGY & MINERALIZATION.....	9
5.2	PROPERTY GEOLOGY.....	10
5.3	PROPERTY MINERALIZATION	10
6	2024 EXPLORATION PROGRAM.....	12
7	SAMPLE PREPARATION, ANALYSIS AND SECURITY	14
8	PROGRAM RESULTS	14
9	DISCUSSION AND CONCLUSIONS	19
10	REFERENCES.....	20

FIGURES

Figure 1.	General location map of the New Sunro Copper property.....	2
Figure 2.	New Sunro Copper Property Claim Map.....	5
Figure 3.	New Sunro Copper Property Geology Map.....	11
Figure 4.	New Sunro Copper Property with rock sample locations and mineralized zones.....	13
Figure 5.	Sample C112331 Gossanous basalt hosting 5% disseminated pyrite, 2% discontinuous pyrite stringers, and rare chalcopyrite.....	15
Figure 6.	Sample C112338 Basalt with strong iron-oxide fracture fill and 5% disseminated pyrite.....	16

Figure 7. Sample C112342 Basalt with 25% semi-massive to disseminated pyrite; slumped subcrop off main blasted face. 17

Figure 8. Sample C112347 Highly oxidized basalt with blebby to semi-massive pyrite and pyrrhotite 18

TABLES

Table 1. Property claims list.....4

Table 2. Yellowcliff Showing assay results table..... 15

Table 3. Tiger Showing assay results table..... 16

Table 4. Winkler Showing assay results table..... 17

Table 5: Pat & Stewart assay results table..... 18

APPENDICES

APPENDIX A	Statement of Expenditures
APPENDIX B	Statement of Qualifications
APPENDIX C	Samples Detail Tables
APPENDIX D	Assay Certificates
APPENDIX E	Analytical Procedure Method Descriptions
APPENDIX F	Sample Maps with Elemental ppm values

1 INTRODUCTION

The New Sunro Copper Project “the Property” in Jordan River is located approximately 50 km west of the city of Victoria B.C. on southern Vancouver Island at a latitude of 48°27’N and longitude 124°03’ W, as shown in Figure 1 below. The property includes a total of 15 mineral claims within the Victoria Mining Division totaling approximately 1393 ha (13.9 km²) and is located within the territories of the Pacheedaht First Nations.

On December 4th and 5th, 2024, Tripoint Geological Services Ltd. managed a small field program sending a crew of 2 geologists to the New Sunro Property to collect rock samples. The main targets of the field program were to locate the Winkler, Tiger, and Yellowcliff showings. In total 18 rock samples were collected, and the total program expenditures were \$11,633.45.

The 2024 field program successfully located the Winkler, Tiger, and Yellowcliff showings along with the Pat and Stewart showings. These mineral occurrences consisted of gossanous basalt to gabbro with disseminated to semi-massive pyrite and minor chalcopyrite and pyrrhotite.

Highlighted results include up to 1.65% Cu, 71.6 ppm Ag, 2560 ppm Ni, 1770 ppm Co, and one sample with up to 145.5 ppm Mo.

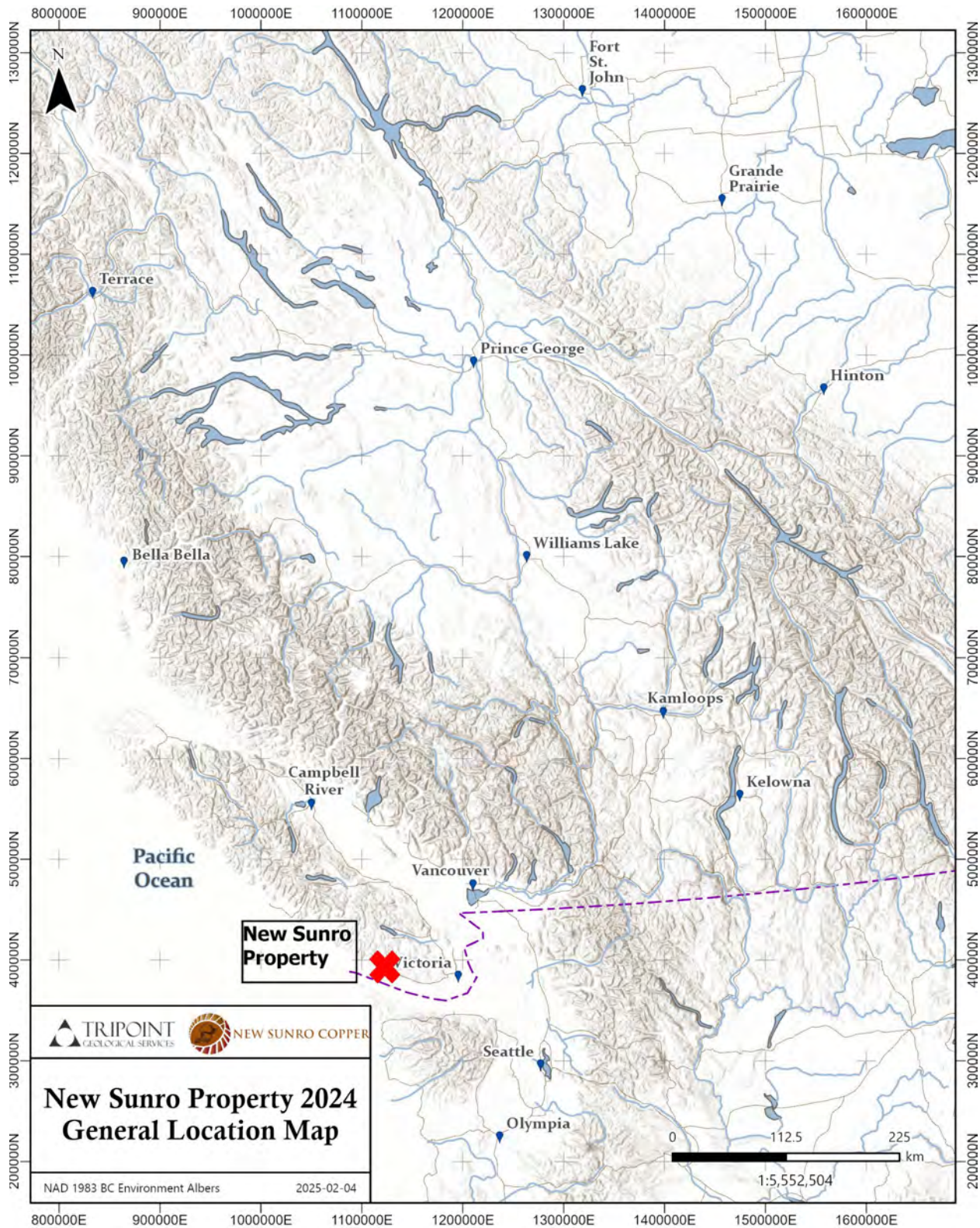


Figure 1. General location map of the New Sunro Copper property.

Prepared by Tripoint Geological Services Ltd.

For **New Sunro Copper Ltd.**

2 LOCATION, ACCESS, PHYSIOGRAPHY & CLIMATE

The New Sunro Copper Property is located on southern Vancouver Island approximately 50 km W from the city of Victoria. The 1393 ha property is made up of 15 contiguous mineral claims on NTS map sheets 092C 08E within the Victoria Mining Division. The claims are centered at 48°26' 49" N and 124° 02' 29" W.

The Property is accessed by taking Highway 14, 25 km west of the city of Victoria through the community of Sooke, and another 25 km west on West Coast Rd to the beginning of Fore Bay East Main logging road. This logging road provides access along the east side of Jordan River to many of the historic mine workings on the east side of the property.

The New Sunro Copper Property is situated on Southern Vancouver Island along the southwest slope of Jordan Ridge where it is bisected by the Jordan River. Topography consists of a rugged, southwest sloping plateau incised by steep sided canyons containing southwest draining creeks. Elevations range from 75 m in the south to 650 m in the north. Most of the Property is covered by second growth mixed forest including active and inactive logging areas. The climate is warm and dry in the summer and cool and very wet in the winter, with little or no snow accumulations but generally heavy rain and winds in the winter. This allows for a year-round field season for most surface and any underground work (Houle, 2023).

3 LAND TENURE AND CLAIM STATUS

The New Sunro Copper Property includes a total of 15 mineral claims totaling approximately 1393 ha (13.9 km²) which are 100% owned by New Sunro Copper. The claim block is currently in good standing. A summary of claims is provided in Table 1.

Table 1. Property claims list.

TENURE NUMBER	CLAIM NAME	OWNER NAME	ISSUED DATE	GOOD TO DATE	AREA (ha)
564267	SUNRO 1	New Sunro Copper	2007/AUG/08	2026/JUN/01	21.4234
663263		New Sunro Copper	2009/NOV/01	2026/JUN/01	107.1172
841971		New Sunro Copper	2010/DEC/30	2026/JUN/01	128.5405
842946		New Sunro Copper	2011/JAN/13	2024/DEC/06	21.4199
953676		New Sunro Copper	2012/MAR/01	2026/JUN/01	42.8538
1016009		New Sunro Copper	2013/JAN/16	2025/SEP/01	107.117
1016282		New Sunro Copper	2013/JAN/24	2026/JUN/01	42.8538
1017854		New Sunro Copper	2013/MAR/17	2025/SEP/01	214.2342
1021332	SUNRO EAST	New Sunro Copper	2013/AUG/01	2025/SEP/01	214.2654
1021619		New Sunro Copper	2013/AUG/12	2025/SEP/01	192.757
1021801		New Sunro Copper	2013/AUG/21	2025/SEP/01	42.8635
1022022		New Sunro Copper	2013/SEP/03	2025/SEP/01	42.8399
1022028		New Sunro Copper	2013/SEP/03	2025/SEP/01	21.42
1022415	SUNRO SOUTH	New Sunro Copper	2013/SEP/17	2025/SEP/01	128.5769
1096947	SUNRO NORTHWEST	New Sunro Copper	2022/AUG/04	2025/SEP/01	64.2599
			Total Area in Hectares		1392.515

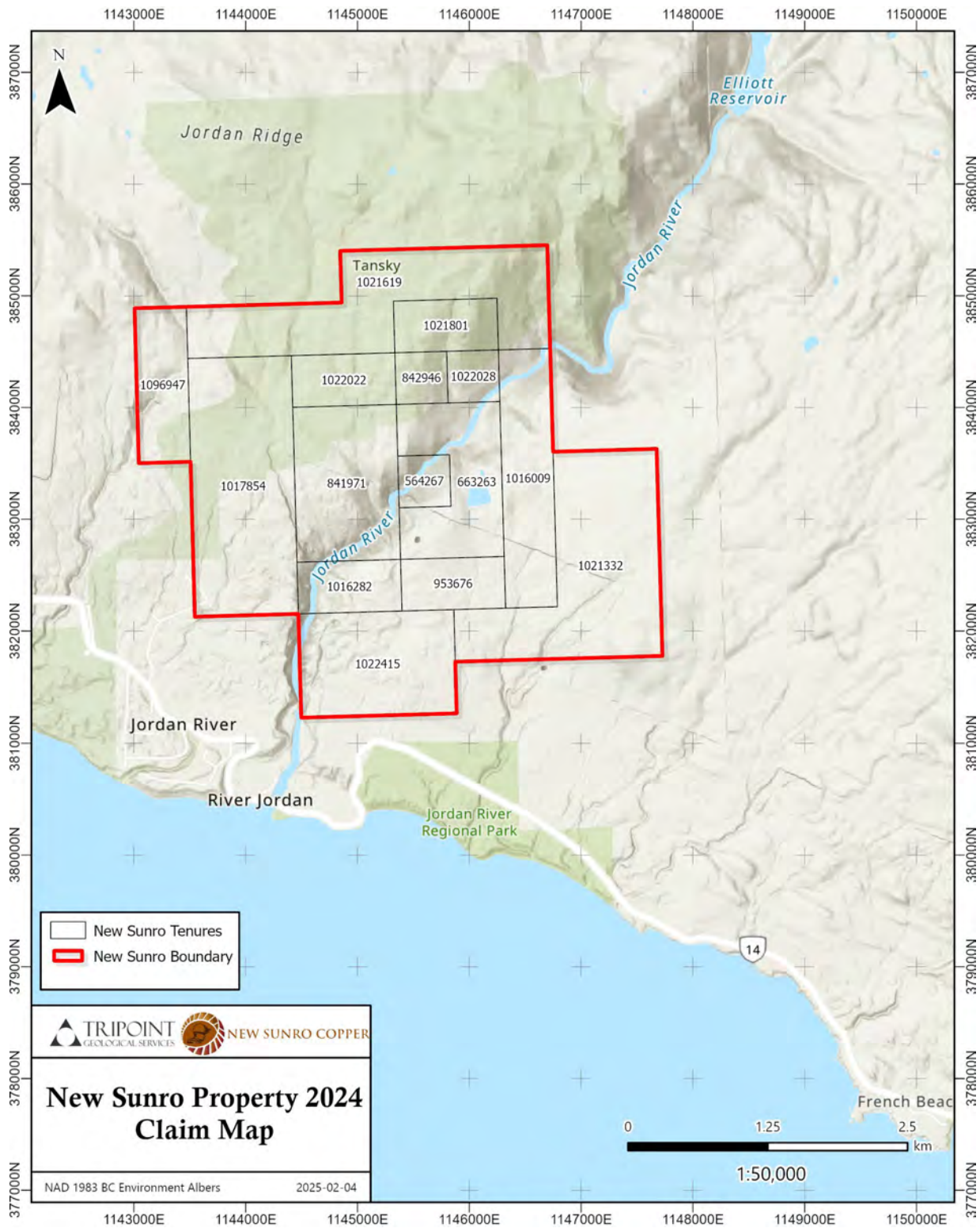


Figure 2. New Sunro Copper Property Claim Map.
 Prepared by Tripoint Geological Services Ltd.
 For **New Sunro Copper Ltd.**

4 EXPLORATION HISTORY

Below is a summary of previous work completed on and around the New Sunro Copper Project.

1958 - Rio Canadian Exploration Ltd. (Gatenby et al., 1958, AR 211)

- Geological mapping, silt and soil samples collected; self-potential geophysics survey completed
- Surface trenching completed on SP survey anomalies
- Further hand trenching done on three prospective trenches
- With soil and silt sampling 6 anomalous areas were found

1963 - Macsan Exploration Ltd. (Malcolm, 1963, AR 544)

- Mapping of mineral claims completed
- Chalcopyrite and pyrrhotite mineralization extensive in sheared and brecciated sections and in hornblende & quartz-plagioclase pegmatite alterations derived from granite

1970 - Eagle Geophysics Ltd. (Lloyd, 1970, AR 2928)

- IP survey completed on Roed project near Jordan River
- Broad anomaly found on west side of grid striking E-W 700 ft wide by 2000 ft long, although most likely broad disseminated sulphides and not like Sunro Type.

1972 - River Jordan Syndicate. (White, 1972, AR 4104)

- Linecutting, magnetometer surveying, electromagnetometer surveying, soil sampling and IP surveying completed on Loss and Wolf claims
- Soil samples yielded within Loss Claims a peak value of 194 ppm Cu and 160 ppm Co, and within the Wolf claims a peak value of 84 ppm Cu and 102 ppm Co.

1977 - Peter A. Boning. (Boning, 1977, AR 6592)

- Examining outcrops where geophysical or geochemical anomalies previously found.
- Geochemical and geophysical anomalies remain unexplained

2003 - Emerald Fields Resource Corp. (Gilmour and McKinley, 2003, AR 27280)

- Geological mapping of logging roads, rock sampling, stream sampling, and VLF-EM magnetometer survey on Pearson PGE and Karen properties
- Copper main metal in skarn showings
- Stream sediment results on Karen property yielded values of gold and copper up 22.4 g/t Au and up to 509 g/t Cu.

2004 - Andris Kakauka. (Kakuaka, 2004, AR 27472)

- 46 soil samples, 6 rock chip samples, and 2.4 km of line surveyed within the Sunloch claim.
- Very high Cu grades in rock chip samples (4367-7942 ppm) with two showings having over 10,000 ppm, River and Cave Zones. Higher Au grades of 720 and 960 ppb in highest grading Cu showings.
- Soil sample results show the highest Cu-Ag-Au anomalies in North portion of grid in River, Center, Turnbull and Cave Zones.

2009 - Explorations Unltd. (Krause, 2009, AR 30545)

- Contour geochemical survey and 24 soil samples collected on DS claims group
- Cu grades in soil up to 63.9 ppm and Zn grades up to 28 ppm

2010 - R. Krause. (Krause, 2010, AR 31362)

- 18 grab samples collected from pit with exposed massive sulphides visibly brecciated with up to 20% quartz with Au and Ag and five one-meter samples collected in situ from pit on DS claims group
- Select rubble and in situ rock samples had Au values up to 7.34gm/mt and three in situ samples had Cu values ranging from 17.86-36.30 ppm.

2011 - M. Miller. (Krause, 2011, AR 31997)

- Geological mapping completed to confirm regional geology and explain geophysical anomalies, and 21 grab samples collected in DS claims group
- Mapping of geology agreed with past surveys of the area

2012 - G. Delourme Diamond Drilling. (Krause, 2012, AR 33480)

- Two diamond drill holes drilled with a total of 241.4 m
- Increase in graphite within tuffaceous felsic rocks but no significant sulphide mineralization

2013 - I. Webster. (Webster, 2013, AR 33931)

- Bedrock exposed on some road cuts and examined, and a stereoscopic air photo interpretation completed on Loss Creek property
- Potential gold bearing fluvial materials located

2016 - Precision GeoSurveys Inc, Auracle Remote Sensing Inc (Houle, 2016, AR 35982)

- Airborne geophysical & digital elevation survey, and magnetic and radiometric airborne surveys completed
- Min zones occur at flexure in Central Gabbro unit and straddle surrounding volcanics with two clusters of geological targets

2022 - SJ Geophysics, J. Houle. (Houle, 2022, AR 40462)

- 3D inversion modelling done from 2015 aeromagnetic survey data, 1.5 km of geological mapping completed, and 7 outcrop and 4 stream samples collected
- High mag-sus areas along N and S contacts of Central Gabbro associated with min zones in center of property
- Soils showed slightly elevated Ag and Cu and three rock samples in Turnbull Zone showed slightly elevated Si, Cu, and V in weakly mineralized mafic volcanics

2023 - J. Houle. (Houle, 2023, AR 41634)

- 3 km of geological mapping, 6 outcrop and 2 stream samples collected
- Outcrop samples in Turnbull Zone yielded 20.9% Cu, 0.552 ppm Au, and 25.2 ppm Ag along with a few other elevated elements
- Highlighted samples include H619236 with 1420 ppm Zi, H619235 with 3910 ppm Cu, and from Hornet Zone specifically sample H619237 with 1280 ppm Cu

5 GEOLOGY

5.1 REGIONAL GEOLOGY & MINERALIZATION

The following geology section is taken from Houle (2023).

The geological setting of the New Sunro Property is simple, with only three major rock units exposed in that portion of the Crescent Terrain, which forms the southernmost and youngest portion of Vancouver Island. Descriptions of the rock units to follow are taken from Geological Survey of Canada publications by Dolmage, V., 1919, Clapp, H.C., 1911 and Clapp, H.C., 1912.

Eocene to Oligocene

Carmanah Group

EOIC undivided sedimentary rocks.

Paleocene to Eocene

Metchosin Igneous Complex

PeEMHgb High-Level Gabbros: gabbroic to dioritic intrusive rocks.

PeEMSDdb Sheeted Dykes: diabase, basaltic intrusive rocks.

PeEMSgb Sooke Gabbro: gabbroic to dioritic intrusive rocks.

PeEMMvb Metchosin Formation: basaltic volcanic rocks.

The oldest rocks are basaltic volcanics of the Metchosin Formation (PeEMMvb), consisting of agglomerates, flow breccias and tuffs, varying from fine to medium-coarse grained, and locally porphyritic. The coarse-grained varieties strongly resemble the gabbroic intrusions. The volcanic flows are locally cut by diabase dykes (PeEMSDdb) of similar composition as the flows and are too small to appear on regional or property scale maps. The intrusive High-Level Gabbro (PeEMHgb) in the New Sunro Property is similar to but more differentiated and altered than the Sooke Gabbro (PeEMSgb) bodies found further east in the Sooke area.

The Carmanah Group (EOIC), known as the Sooke Formation in older reports and maps, consists of partly consolidated, finely bedded sediments unconformably overlying the older rocks of the Metchosin Igneous Complex.

Shearing within the contact metamorphosed volcanics is associated with the most intense hornblende alteration accompanied by metallic minerals magnetite, pyrite, pyrrhotite and chalcopyrite, and cut by veinlets and irregular replacements of quartz, epidote, zoisite, chlorite, calcite, secondary feldspar and sulphides.

5.2 PROPERTY GEOLOGY

From (Houle, 2023). See Figure 3.

In the Property area, the gabbro is black, irregularly grained and consists mainly of hornblende and feldspar, with minor augite, ilmenite, magnetite and titanite. Masses of gabbro intrude the Metchosin volcanics in steeply dipping, elongate, dyke-like bodies parallel to the west-northwest fold axis in the volcanics. Three major gabbro intrusive bodies cross the Property, each about 1 kilometre apart and each approximately 250 to 750 metres thick, with the thickest dyke crossing the centre of the Property. Shallow inliers of Sooke Formation occur in the southwest and southeast areas of the New Sunro Property.

5.3 PROPERTY MINERALIZATION

From (Houle, 2023).

The Property hosts Eocene age metallic mineralization consisting of clustered, copper-gold silver bearing, steeply dipping sulphide gash veins and lenticular masses, classified as Tholeiitic Intrusion Hosted Ni-Cu (BCGS Mineral Deposit Profile M02) assigned to the past producer BC MINFILE 092C073 – Sunro. Recently developed USGS Mineral Deposits Models have created a broad classification of magmatic sulfide-rich nickel-copper deposits including the sub-class: Basalt-related dike-sill hosted Ni-Cu Deposits, which includes the giant class mineral deposit Noril'sk-Talnakh, and also describes the form and mode of occurrence of the copper-silver-gold bearing zones on the Property.

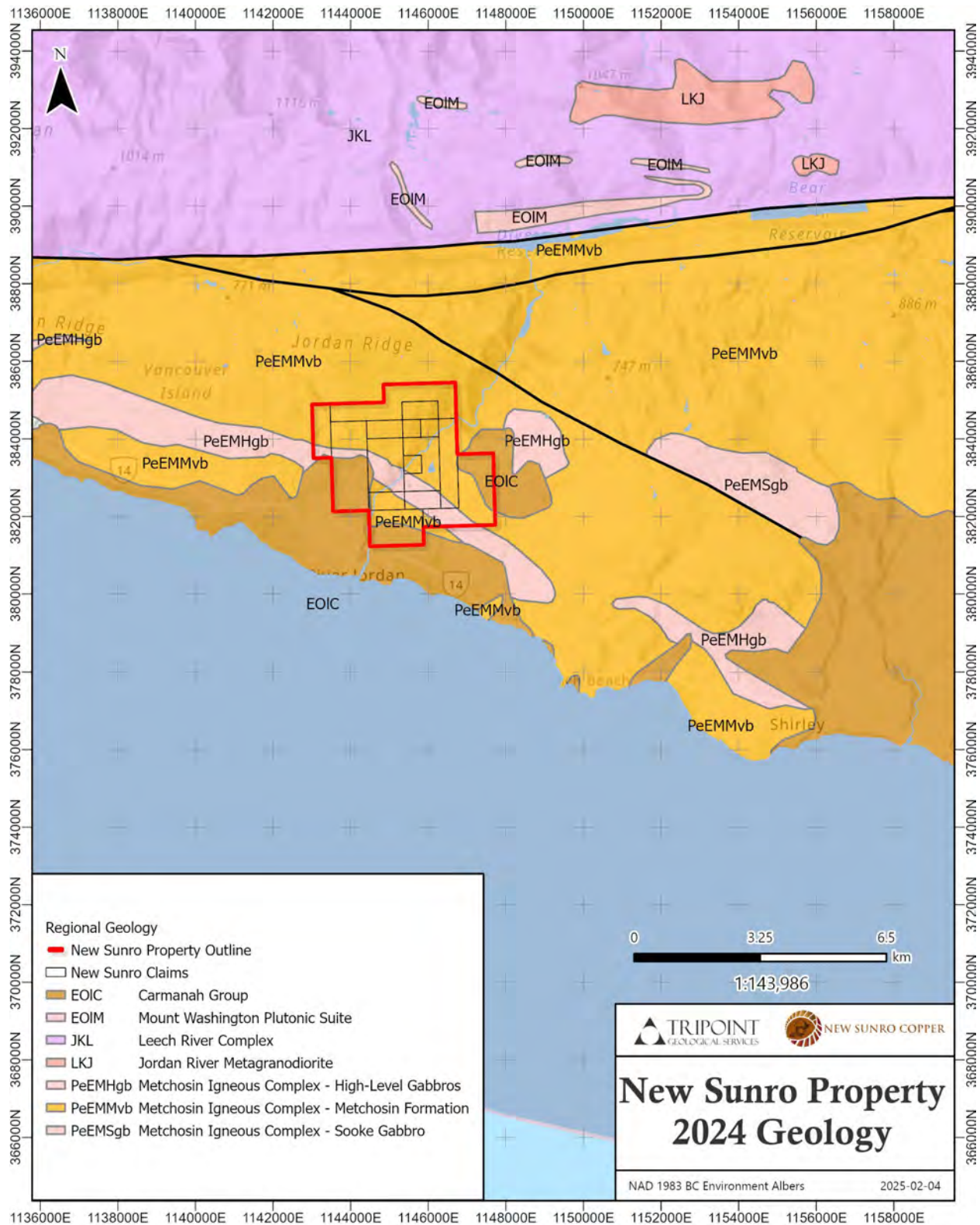


Figure 3. New Sunro Copper Property Geology Map

Prepared by Tripoint Geological Services Ltd.

For **New Sunro Copper Ltd.**

6 2024 EXPLORATION PROGRAM

The 2024 Exploration program included a total of 18 rock samples collected by two Tripoint geologists from the New Sunro Property in Jordan River on Dec 4th and Dec 5th, 2024. All relevant rock sampling information was inputted into ESRI Field Maps on tablets, and photos were taken of each respective sample by Tripoint personnel.

The main targets of the field program were to locate the Winkler, Tiger, and Yellowcliff showings. Below in Figure 4 is a detailed map displaying target mineral showings and samples collected.

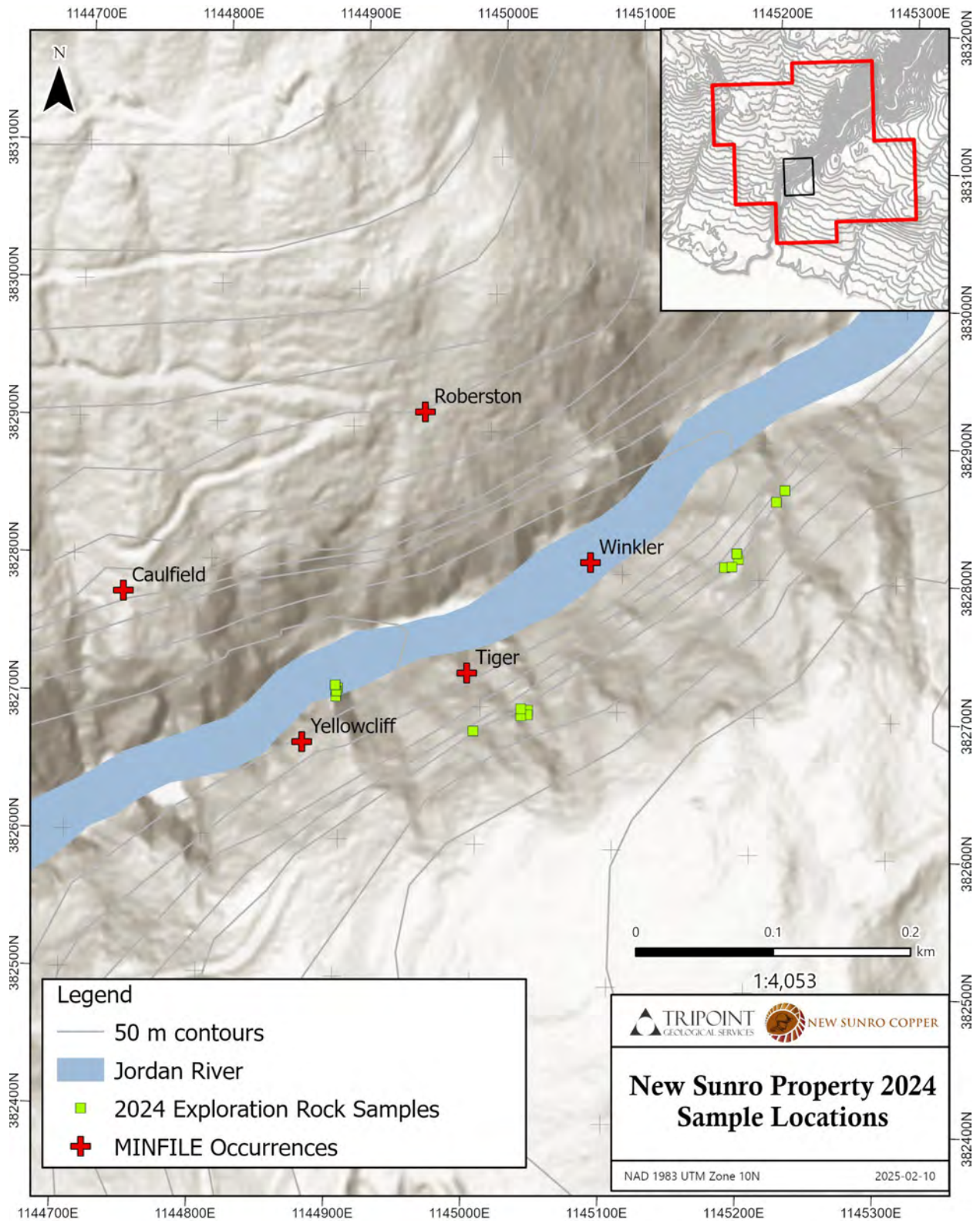


Figure 4. New Sunro Copper Property with rock sample locations and mineralized zones.

Prepared by Tripoint Geological Services Ltd.

For **New Sunro Copper Ltd.**

7 SAMPLE PREPARATION, ANALYSIS AND SECURITY

All rock sample data was collected on field tablets using the ESRI Field Maps application. At each sampling site, 640 – 1600 g of rock material was placed in a poly bag with their associated sample tag and labelled with the sample tag number. Sample locations were marked using metal butter tags and flagging tape labelled with sample tag number and year. Sample coordinates were recorded on a handheld Garmin GPS for redundancy. Samples were securely packaged into rice bags, fastened with security tags, and shipped to ALS Kamloops for sample preparation where they were later transferred to ALS Vancouver for assays and analyses.

Sample preparation (PREP-31A) was carried out at ALS North Vancouver, where samples were dried and crushed until at least 70% passes a 2 mm (Tyler 10 mesh) screen. A split of up to 250 g was taken using a riffle splitter, pulverized until at least 85% passes a 75-micron (Tyler 200 mesh) screen and sent for analysis.

Sample analysis was carried out at ALS Vancouver, where each sample was sent for two tests. For Au values (Au-AA23), 30 g of a prepared sample was fused with a mixture of lead oxide, sodium carbonate, borax, silica, and other reagents as required, inquarted with 6 mg of gold-free silver and then cupelled to yield a metal bead. The bead is digested in 0.5 mL of dilute nitric acid in a microwave oven; 0.5 mL concentrated hydrochloric acid is then added and further digested at a lower power setting. The digested solution is cooled, diluted to a total volume of 4 mL with de-mineralized water, and analyzed by atomic absorption spectroscopy against matrix-matched standards. For non-Au multi-element values (ME-MS61), 0.25 g of a prepared sample undergoes a four-acid digestion, starting with a combination of nitric, perchloric, and hydrofluoric acid followed by a final dissolution stage using hydrochloric acid. The digested sample is then analyzed using a combination of ICP-MS and ICP-AES for trace-level elements.

8 PROGRAM RESULTS

The 2024 field program on the New Sunro Copper Property overall was successful in locating several known mineral occurrences. These mineral occurrences all consisted of gossanous basalt to gabbro with disseminated to semi-massive pyrite and minor pyrrhotite and chalcopyrite. Each of these showings were found to follow parallel 5 cm to 3 m wide shear zones which were dipping west very steeply. Many of these structures were in the blasted face of the abandoned railway.

Below are descriptions of each mineral occurrence (showing) with associated assay results and analysis.

Yellowcliff Showing:

A total of four (4) rock samples were collected from the Yellowcliff showing, from both outcrop and the debris below. Samples consisted of gossanous basalt with roughly 5-10% disseminated pyrite with some minor pyrite stringers. One sample contained trace chalcopyrite.

Sample results show three samples with >1% Cu. **C112329** with 1.1%, **C112330** with 1.47% and **C112331** with 1.39%. Associated with these high Cu values are slightly elevated values of Ag, Ni, and Co. Sample C112331 interestingly has a slightly elevated value of Mo. Assay results listed below (Table 2) and Sample C112331 shown below (Figure 5).

Table 2. Yellowcliff Showing assay results table.

SAMPLE NUMBER	SAMPLE TYPE	Au (ppm)	Cu (ppm)	Ag (ppm)	Mo (ppm)	Ni (ppm)	Co (ppm)
C112329	Rock	0.312	11000	11.15	0.34	22	70.7
C112330	Rock	0.247	14700	8.8	0.24	18.2	93.8
C112331	Rock	0.364	13850	12.95	4.85	20.7	81
C112332	Rock	0.061	1845	2.76	1.5	33.6	63.8



Figure 5. Sample C112331 Gossanous basalt hosting 5% disseminated pyrite, 2% discontinuous pyrite stringers, and rare chalcopyrite.

Tiger Showing:

A total of four (4) rock samples were collected from Tiger showing from mineralized material which was cut by an abandoned railway. Samples consisted of gossanous basalt with roughly 5-20% disseminated to semi-massive pyrite and pyrite stringers.

Sample results show two samples with >0.5% Cu. **C112334** with 0.81% Cu and **C1122338** with 0.83% Cu. Associated with these high Cu values are slightly elevated values of Ag, Ni, and Co. In sample C112335 there is 0.26% Cu with very elevated levels of Ni and Co, 0.26% and 0.18% respectively. Assay results listed below (Table 3) and sample C112338 shown below (Figure 6).

Table 3. Tiger Showing assay results table.

SAMPLE NUMBER	SAMPLE TYPE	Au (ppm)	Cu (ppm)	Ag (ppm)	Mo (ppm)	Ni (ppm)	Co (ppm)
C112334	Rock	0.065	8100	2.06	0.15	68.5	96.7
C112335	Rock	0.034	2620	0.82	0.26	2560	1770
C112336	Rock	0.097	3640	2.83	1.44	79.3	95.1
C112338	Rock	0.085	8300	2.88	0.61	86.9	100



Figure 6. Sample C112338 Basalt with strong iron-oxide fracture fill and 5% disseminated pyrite.

Winkler Showing:

A total of four (4) rock samples were collected from mineralized material which was cut by an abandoned railway. Samples consisted of gossanous basalt with roughly 5-20% disseminated to semi-massive pyrite and pyrite stringers.

Sample results show two samples with >0.5% Cu. C112342 with 1.43% and C112343 with 0.67%. Associated with these high Cu values are moderately elevated values of Ni and Co, with slightly to moderately elevated Ag values. Sample C112343 has very elevated Mo value compared to the other three samples, which is quite interesting as Mo is not mentioned in past sampling programs. Assay results listed below (Table 4) and sample C112342 shown below (Figure 7).

Table 4. Winkler Showing assay results table.

SAMPLE NUMBER	SAMPLE TYPE	Au (ppm)	Cu (ppm)	Ag (ppm)	Mo (ppm)	Ni (ppm)	Co (ppm)
C112340	Rock	0.039	2330	1.07	0.12	98.9	87.2
C112341	Rock	0.093	840	2.02	0.37	117	68.3
C112342	Rock	0.238	14250	6.53	0.18	494	268
C112343	Rock	0.742	6680	71.6	145.5	264	122

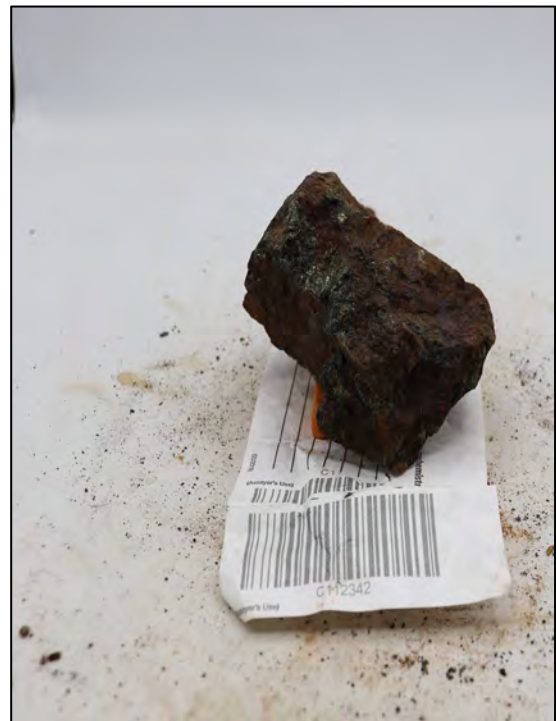


Figure 7. Sample C112342 Basalt with 25% semi-massive to disseminated pyrite; slumped subcrop off main blasted face.

Pat & Stewart Showings:

A total of three (3) rock samples were collected, two from Pat Showing and one sample from the Stewart showing. The samples in the Pat showing consisted of gossanous basalt with strong FeOx staining visible on fractures faces and mineralization of 5% disseminated pyrite and pyrrhotite. The sample taken in the Stewart showing was from a 3 x 2 m outcrop that consisted of coarse-grained gabbro with strong FeOx staining and roughly 5% disseminated pyrite.

Sample results show two samples with >0.8% Cu. C112346 with 0.85% and C112347 with 1.65%. Associated with these high Cu values are slightly elevated Ag, and weakly-moderately elevated Ni and Co values. Assay results listed below (Table 5) and sample C112347 shown below (Figure 8).

Table 5: Pat & Stewart assay results table.

SAMPLE NUMBER	SAMPLE TYPE	Au (ppm)	Cu (ppm)	Ag (ppm)	Mo (ppm)	Ni (ppm)	Co (ppm)
C112346	Rock	0.107	8530	3.24	0.36	167	86.2
C112347	Rock	0.357	16450	6.33	0.3	752	243
D704129	Rock	-0.005	996	0.34	0.65	311	96.7

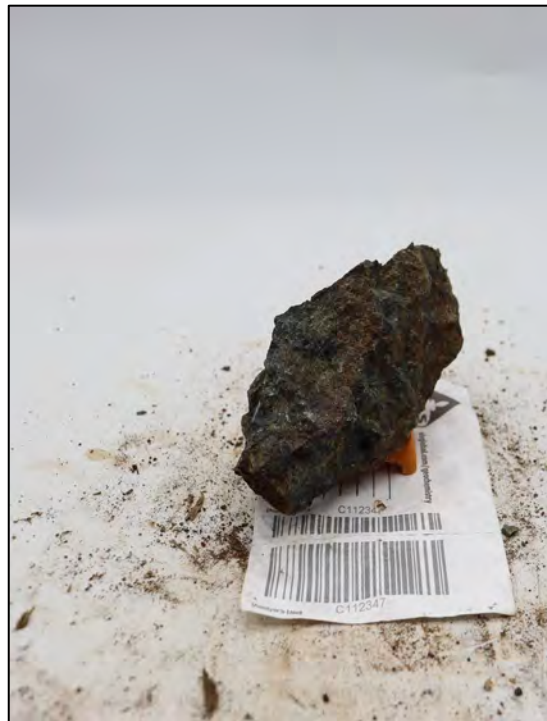


Figure 8. Sample C112347 Highly oxidized basalt with blebby to semi-massive pyrite and pyrrhotite

9 DISCUSSION AND CONCLUSIONS

The 2024 field program was successful in locating several known mineral occurrences. While observing these occurrences it was found that each of these showings paralleled 5 cm to 3 cm wide shear zones which were very steeply dipping to the west. Within these mineral occurrences, high grade Cu values were found ranging from 0.67-1.65% and were often found to be associated with moderately elevated Ni and Co and slightly elevated Ag levels.

The average grades for Cu, Au, and Ag at the historically producing Sunro Mine were 1.03%, 0.153 g/t, and 1.70 g/t respectively (BC MINFILE 092C 073). Looking at the sample grades in the results tables above in results section, it is reasonable when compared to average grades of samples collected.

The current geological rock types are consistent with previous observation of the New Sunro property, with similar pyrite-pyrrhotite and chalcopyrite mineralization found in disseminated form found near gabbro and feldspathic stringers in areas of intense shearing (Gordon, T.M. and Malone, A.S., 1970, BC ARIS File 2229).

Future exploration programs should focus on tracing these mineralized structures along strike. This would be done by continued rock sampling where outcrop allows. Due to the very steep nature of the banks of Jordan River, it is recommended that rope scaling techniques be used to trace the showings west below the abandoned railway.

A conventional soil sample program should be conducted to trace the structures along strike above the railway. Exploration work completed during the 2004 program (ARIS 27472) shows that the soil here has a good correlation between mineralization and copper-in-soil grades.

This proposed 300-soil program with 50 m spacing is recommended over the showing and ground to the east. The goal of this proposed rock and soil program would be to identify drill targets along the Yellowcliff, Tiger, Winkler, Stewart, and Pat structures

10 REFERENCES

- Boning, P.A. (1977) Geology of the Joan Claims Areas on Jordan Ridge Victoria Mining Division British Columbia. Assessment Report **6592**.
- Clapp, C.H. (1911) Geological Reconnaissance Southern Vancouver Island Map 17A to accompany Memoir No. 13, Geological Survey of Canada.
- Clapp, C.H. (1912) Southern Vancouver Island, Memoir No. 13, Geological Survey of Canada.
- Dolmage, V. (1920) Sunloch Copper District, BC, in GSC Summary Report 1919, Part B, p.20-30.
- Gatenby et al. (1958) Geological, Geochemical and Geophysical Report on the Ann, Don, Jo A, Jo B, Jo C and Jo D group of mineral claims in the Victoria Mining division 4 miles NW of Jordan River. Assessment Report **211**.
- Gilmour, W.R., and McKinley, S.D. (2003) Assessment Report on the Pearson PGE Property. Assessment Report **27280**.
- Gordon, T.M. and Malone, A.S. (1970) Geological and Geochemical Report on the Mead Claim Group, Victoria Mining Division, for Quintana Minerals Corporation. BC ARIS File 2229
- Houle, J. (2016) 2015 Technical Assessment Report for the New Sunro Property Vancouver Island, British Columbia. Assessment Report **35982**.
- Houle, J. (2022) 2022 Assessment Report for Geophysical, Geological, Geochemical & Prospecting on the Jordan River Property Vancouver Island, British Columbia. Assessment Report **40462**.
- Houle, J. (2023) 2023 Assessment Report for Geological, Geochemical, Prospecting & Preparatory Surveys on the Jordan River Property Vancouver Island, British Columbia. Assessment Report **41634**.
- Kakuaka, A., (2004) Geological & Geochemical Report on the Sunloch Claim (Sunro Copper-Silver-Gold Mine) Jordan River, B.C. Assessment Report **27472**.
- Krause, R. (2009) Geochemical Assessment Report on the DS Claim Group In the Victoria Mining District. Assessment Report **30545**.

- Krause, R. (2010) Geochemical and Prospecting Assessment Report on the 2009 Exploration Program on the DS Claim Group Victoria Mining Division Jordan River Area Vancouver Island. Assessment Report **31362**.
- Krause, R. (2011) Geological Mapping Assessment Report on the 2010 Exploration Program on the DS Claim Group Victoria Mining Division Jordan River Area Vancouver Island. Assessment Report **31997**.
- Krause, R. (2012) Diamond Drilling Assessment Report on the 2011 Exploration Program on the DS Claim Group Victoria Mining Division Jordan River Area Vancouver Island. Assessment Report **33480**.
- Lloyd, J. (1970) A Geophysical Report on a time domain induced polarization survey on the Roed Copper Project, Jordan River, Vancouver Island. Assessment Report **2928**.
- Malcolm, D.C. (1963) EXT Group Geological Report. Assessment Report **544**.
- Webster, I. (2013) Geological Air Photo Interpretation of the Loss Creek Placer Property. Assessment Report **33931**.
- White, G.E. (1972) Geophysical Geochemical Report River Jordan Syndicate on Loss and Wolf Claims. Assessment Report **4104**.

APPENDIX A

STATEMENT OF EXPENDITURES

New Sunro Copper Ltd. - 2024 Jordan River Project Statement of Costs						Totals
Personnel Expenditures						
Project Planning						
Personnel (Name)	Details	Hours	Rate	GST Subtotal*		
Michael Brinton		5.67	\$105.00	\$29.77	\$625.12	
Bradley Galloway		6.17	\$105.00	\$32.39	\$680.24	
					\$1,305.36	
\$1,305.36						
Field Work						
Personnel (Name) * / Position	Field Days (list actual days)	Days	Rate	GST Subtotal*		
Michael Brinton/Senior Geologist	Dec 4-5	2	\$1,050.00	\$105.00	\$2,205.00	
Matthew Temple/Exploration Geologist	Dec 4-5	2	\$780.00	\$78.00	\$1,638.00	
					\$3,843.00	
\$3,843.00						
Post Field Data Organization						
Personnel (Name)	Details	Hours	Rate	GST Subtotal*		
Michael Brinton		6	\$105.00	\$31.50	\$661.50	
Bradley Galloway		1	\$105.00	\$5.25	\$110.25	
					\$771.75	
\$771.75						
Report Writing						
Personnel (Name)	Details	Hours	Rate	GST Subtotal*		
Matthew Temple		36.23	\$78.00	\$141.30	\$2,967.24	
Samantha Super		1.5	\$55.00	\$4.13	\$86.63	
Michael Brinton		4	\$105.00	\$21.00	\$441.00	
					\$3,494.86	
\$3,494.86						
Other Expenditures						
Geochemical Surveying	Number of Samples	No.	Rate	Subtotal		
Rock Samples	20 Rock Samples	20.0	\$71.43	\$1,428.60		
Other (specify)	XRF Analysis		\$0.00	\$245.70		
					\$1,674.30	
\$1,674.30						
Transportation* Includes GST	Details	Subtotal				
Mileage		\$269.23				
					\$269.23	
\$269.23						
Accommodation & Food* Incl. GST	Rates per day	Days	Rate	Subtotal		
Meals	Actual Costs			\$143.88		
					\$143.88	
\$143.88						
Freight *Includes GST	Details	Subtotal				
Shipping		\$131.07				
					\$131.07	
\$131.07						
TOTAL Expenditures						
\$11,633.45						

APPENDIX B

STATEMENT OF QUALIFICATIONS

I, Matthew, do hereby declare that:

- 1) I am the author of the report titled "Assessment Report on the New Sunro Copper Project 2024", relating to the New Sunro Copper Property.
- 2) At the time of writing this report I was an Exploration Geologist of Tripoint Geological Services Ltd. of 148-2770 Leigh Road Langford BC V9B 4G1;
- 3) I hold a Bachelor of Science (B.Sc.) in Earth Sciences granted by the University of Victoria in 2020;
- 4) I am a Geologist In Training with the Engineers and Geoscientists British Columbia (EGBC), Member Number #242409;
- 5) I have been involved in the mineral exploration of gold, silver, copper, and nickel in Canada and the United States since 2019; and
- 6) I am not aware of any material fact or material change with respect to the subject matter of this report.

Dated at Victoria, British Columbia with effective date of February 19, 2025.



Matthew Temple, B.Sc., G.I.T

APPENDIX C

SAMPLE DETAIL TABLES

[illegible]

APPENDIX D

ASSAY CERTIFICATES



ALS Canada Ltd.
2103 Dollarton Hwy
North Vancouver BC V7H 0A7
Phone: +1 604 984 0221 Fax: +1 604 984 0218
www.alsglobal.com/geochemistry

To: TRIPOINT GEOLOGICAL SERVICES
2770 LEIGH RD #148
VICTORIA BC V9B 4G1

Page: 1
Total # Pages: 2 (A – D)
Plus Appendix Pages
Finalized Date: 5-JAN-2025
This copy reported on 7-FEB-2025
Account: TRIPGEO

CERTIFICATE VA24357365

Project: New Sunro
P.O. No.: SUN24-001
This report is for 20 samples of Rock submitted to our lab in Vancouver, BC, Canada on 11-DEC-2024.

The following have access to data associated with this certificate:

TRIPOINT ASSAYS

SAMPLE PREPARATION

ALS CODE	DESCRIPTION
WEI-21	Received Sample Weight
LOG-21	Sample logging – ClientBarCode
LOG-23	Pulp Login – Rcvd with Barcode
CRU-QC	Crushing QC Test
PUL-QC	Pulverizing QC Test
CRU-31	Fine crushing – 70% <2mm
SPL-21	Split sample – riffle splitter
PUL-31	Pulverize up to 250g 85% <75 um
DISP-01	Disposal of all sample fractions

ANALYTICAL PROCEDURES

ALS CODE	DESCRIPTION	INSTRUMENT
ME-MS61	48 element four acid ICP-MS	
ME-OG62	Ore Grade Elements – Four Acid	ICP-AES
Cu-OG62	Ore Grade Cu – Four Acid	
Au-AA23	Au 30g FA-AA finish	AAS

This is the Final Report and supersedes any preliminary report with this certificate number. Results apply to samples as submitted. All pages of this report have been checked and approved for release.

***** See Appendix Page for comments regarding this certificate *****

Signature:

Saa Traxler, Director, North Vancouver Operations

***** See Appendix Page for comments regarding this certificate *****



ALS Canada Ltd.
2103 Dollarton Hwy
North Vancouver BC V7H 0A7
Phone: +1 604 984 0221 Fax: +1 604 984 0218
www.alsglobal.com/geochemistry

To: TRIPOINT GEOLOGICAL SERVICES
2770 LEIGH RD #148
VICTORIA BC V9B 4G1

Page: 2 – C
Total # Pages: 2 (A – D)
Plus Appendix Pages
Finalized Date: 5-JAN-2025
Account: TRIPGEO

Project: New Sunro

CERTIFICATE OF ANALYSIS VA24357365

Sample Description	Method Analyte Units LOD	ME-MS61	ME-MS61	ME-MS61	ME-MS61	ME-MS61	ME-MS61	ME-MS61	ME-MS61	ME-MS61	ME-MS61	ME-MS61	ME-MS61	ME-MS61	ME-MS61
		Pb	Rb	Re	S	Sb	Sc	Se	Sn	Sr	Ta	Te	Th	Ti	Tl
		ppm	ppm	ppm	%	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	%	ppm
		0.5	0.1	0.002	0.01	0.05	0.1	1	0.2	0.2	0.05	0.05	0.01	0.005	0.02
C112329		1.9	0.9	0.018	1.56	0.09	30.4	17	1.4	30.1	0.33	0.99	0.56	0.635	<0.02
C112330		<0.5	1.2	0.036	2.13	0.06	30.2	22	1.2	33.7	0.42	0.76	0.36	0.835	<0.02
C112331		1.8	0.7	0.035	2.13	0.06	23.6	18	1.0	47.4	0.25	1.01	0.45	0.513	<0.02
C112332		0.5	1.4	0.075	0.18	0.07	37.2	2	1.5	64.0	0.54	0.23	0.91	0.835	<0.02
C112333		<0.5	1.2	0.002	0.21	0.06	41.6	<1	0.7	148.0	0.44	0.08	0.63	0.936	<0.02
C112334		<0.5	0.9	<0.002	0.92	0.05	25.7	3	1.2	53.5	0.29	0.24	0.52	0.622	<0.02
C112335		0.6	0.3	0.034	>10.0	0.05	2.3	80	0.6	31.8	<0.05	1.33	0.25	0.140	<0.02
C112336		0.6	1.0	0.007	1.14	<0.05	19.3	9	1.1	51.8	0.19	0.49	0.38	0.420	<0.02
C112338		0.6	0.6	0.005	0.87	<0.05	23.6	7	1.1	71.2	0.21	0.31	0.58	0.472	<0.02
C112340		<0.5	1.8	<0.002	0.35	0.26	39.6	1	1.3	29.0	0.17	0.06	0.40	0.390	<0.02
C112341		0.5	2.5	0.003	0.43	0.10	22.3	6	1.0	105.5	0.23	0.23	0.49	0.451	<0.02
C112342		0.8	0.8	0.019	3.52	0.07	25.5	14	1.3	10.0	0.18	0.70	0.48	0.336	<0.02
C112343		5.3	0.2	0.048	0.59	0.14	12.2	175	1.3	164.0	0.16	7.77	2.72	0.261	<0.02
C112344		0.6	2.2	<0.002	0.10	0.05	35.2	1	0.7	52.2	0.07	0.08	0.11	0.302	<0.02
C112345		0.8	2.4	<0.002	0.10	0.05	42.3	<1	0.6	94.3	0.12	<0.05	0.19	0.374	<0.02
C112346		<0.5	1.5	<0.002	1.21	0.09	41.0	5	0.6	133.5	0.13	0.20	0.10	0.673	<0.02
C112347		0.7	2.0	0.003	5.28	0.09	31.5	24	0.4	83.4	0.06	0.69	0.08	0.238	0.02
D704129		0.5	0.7	0.015	1.17	<0.05	8.9	15	1.2	59.7	<0.05	0.21	0.22	0.151	<0.02
D704130		1055	54.6	0.003	3.80	6.56	11.9	3	9.1	98.5	0.82	0.14	8.33	0.257	5.16
D704131		5.3	36.7	<0.002	0.07	0.35	6.2	<1	2.3	197.0	0.40	<0.05	3.14	0.186	0.16

***** See Appendix Page for comments regarding this certificate *****



ALS Canada Ltd.
2103 Dollarton Hwy
North Vancouver BC V7H 0A7
Phone: +1 604 984 0221 Fax: +1 604 984 0218
www.alsglobal.com/geochemistry

To: **TRIPOINT GEOLOGICAL SERVICES**
2770 LEIGH RD #148
VICTORIA BC V9B 4G1

Page: Appendix 1
Total # Appendix Pages: 1
Finalized Date: 5-JAN-2025
Account: TRIPGEO

Project: New Sunro

CERTIFICATE OF ANALYSIS VA24357365
--

	CERTIFICATE COMMENTS
--	-----------------------------

	ANALYTICAL COMMENTS
--	----------------------------

Applies to Method:	REEs may not be totally soluble in this method. ME-MS61
--------------------	--

	LABORATORY ADDRESSES
--	-----------------------------

Applies to Method:	Processed at ALS Vancouver located at 2103 Dollarton Hwy, North Vancouver, BC, Canada.			
	Au-AA23	CRU-31	CRU-QC	Cu-OG62
	DISP-01	LOG-21	LOG-23	ME-MS61
	ME-OG62	PUL-31	PUL-QC	SPL-21
	WEI-21			

APPENDIX E

ANALYTICAL PROCEDURE AND METHOD DESCRIPTIONS



Sample Preparation Package

PREP-31

Standard Sample Preparation: Dry, Crush, Split and Pulverize

Sample preparation is the most critical step in the entire laboratory operation. The purpose of preparation is to produce a homogeneous analytical sub-sample that is fully representative of the material submitted to the laboratory.

The sample is logged in the tracking system, weighed, dried and finely crushed to better than 70 % passing a 2 mm (Tyler 9 mesh, US Std. No.10) screen. A split of up to 250 g is taken and pulverized to better than 85 % passing a 75 micron (Tyler 200 mesh, US Std. No. 200) screen. This method is appropriate for rock chip or drill samples.

Method Code	Description
LOG-22	Sample is logged in tracking system and a bar code label is attached.
CRU-31	Fine crushing of rock chip and drill samples to better than 70 % of the sample passing 2 mm.
SPL-21	Split sample using riffle splitter.
PUL-31	A sample split of up to 250 g is pulverized to better than 85 % of the sample passing 75 microns.

Revision 03.03
March 29, 2012

RIGHT SOLUTIONS RIGHT PARTNER

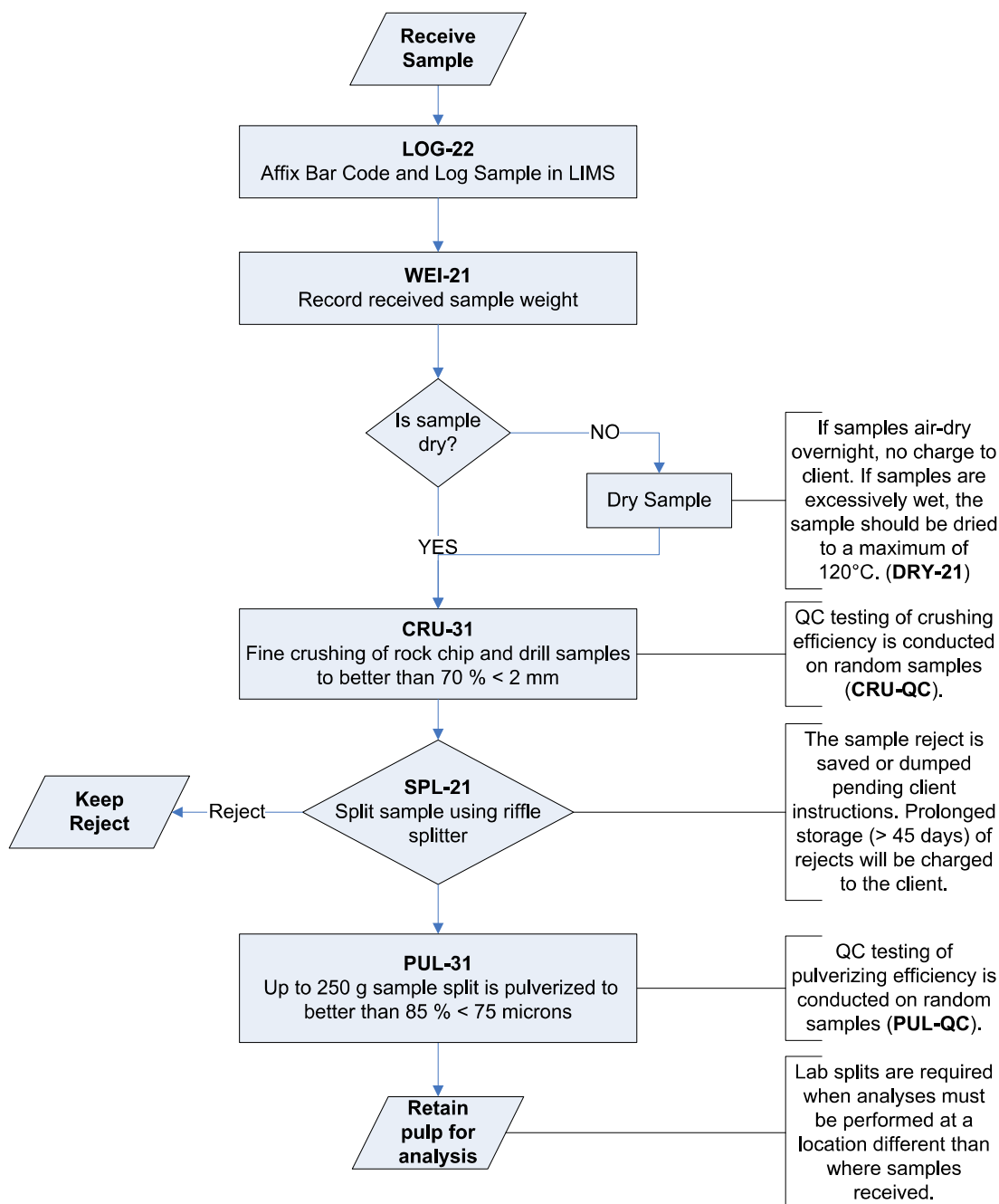
www.alsglobal.com



Sample Preparation Package

Flow Chart -

Sample Preparation Package – PREP-31 Standard Sample Preparation: Dry, Crush, Split and Pulverize



Revision 03.03
March 29, 2012

RIGHT SOLUTIONS RIGHT PARTNER

www.alsglobal.com

Au-AA23 & Au-AA24 – Fire Assay Fusion, AAS Finish

Sample Decomposition:

Fire Assay Fusion (FA-FUS01 & FA-FUS02)

Analytical Method:

Atomic Absorption Spectroscopy (AAS)

A prepared sample is fused with a mixture of lead oxide, sodium carbonate, borax, silica and other reagents as required, inquarted with 6 mg of gold-free silver and then cupelled to yield a precious metal bead.

The bead is digested in 0.5 mL dilute nitric acid in the microwave oven, 0.5 mL concentrated hydrochloric acid is then added and the bead is further digested in the microwave at a lower power setting. The digested solution is cooled, diluted to a total volume of 4 mL with de-mineralized water, and analyzed by atomic absorption spectroscopy against matrix-matched standards.

List of Reportable Analytes:

Method Code	Element	Symbol	Units	Sample Weight (g)	Lower Limit	Upper Limit	Default Overlimit Method
Au-AA23	Gold	Au	ppm	30	0.005	10.0	Au-GRA21
Au-AA24	Gold	Au	ppm	50	0.005	10.0	Au-GRA22

ME-MS61: Ultra-Trace Level Method Using ICP MS and ICP-AES

Sample Decomposition:

HF-HNO₃-HClO₄ acid digestion, HCl leach (GEO-4A01)

Analytical Method:

Inductively Coupled Plasma - Atomic Emission Spectroscopy (ICP-AES)

Inductively Coupled Plasma - Mass Spectrometry (ICP-MS)

The ME-MS61 Ultra Trace method combines a four-acid digestion with ICP-MS instrumentation. A four acid digestion quantitatively dissolves nearly all minerals in the majority of geological materials.

A prepared sample (0.25 g) is digested with perchloric, nitric and hydrofluoric acids. The residue is leached with dilute hydrochloric acid and diluted to volume.

The final solution is then analyzed by inductively coupled plasma-atomic emission spectrometry and inductively coupled plasma-mass spectrometry. Results are corrected for spectral inter-element interferences.

List of Reportable Analytes:

Analyte	Symbol	Units	Lower Limit	Upper Limit
Silver	Ag	ppm	0.01	100
Aluminum	Al	%	0.01	50
Arsenic	As	ppm	0.2	10000
Barium	Ba	ppm	10	10000
Beryllium	Be	ppm	0.05	1000
Bismuth	Bi	ppm	0.01	10000
Calcium	Ca	%	0.01	50
Cadmium	Cd	ppm	0.02	1000
Cerium	Ce	ppm	0.01	500
Cobalt	Co	ppm	0.1	10000
Chromium	Cr	ppm	1	10000
Cesium	Cs	ppm	0.05	500
Copper	Cu	ppm	0.2	10000
Iron	Fe	%	0.01	50
Gallium	Ga	ppm	0.05	10000
Germanium	Ge	ppm	0.05	500
Hafnium	Hf	ppm	0.1	500
Indium	In	ppm	0.005	500
Potassium	K	%	0.01	10
Lanthanum	La	ppm	0.5	10000
Lithium	Li	ppm	0.2	10000
Magnesium	Mg	%	0.01	50
Manganese	Mn	ppm	5	100000
Molybdenum	Mo	ppm	0.05	10000
Sodium	Na	%	0.01	10
Niobium	Nb	ppm	0.1	500
Nickel	Ni	ppm	0.2	10000

Analyte	Symbol	Units	Lower Limit	Upper Limit
Phosphorous	P	ppm	10	10000
Lead	Pb	ppm	0.5	10000
Rubidium	Rb	ppm	0.1	10000
Rhenium	Re	ppm	0.002	50
Sulphur	S	%	0.01	10
Antimony	Sb	ppm	0.05	10000
Scandium	Sc	ppm	0.1	10000
Selenium	Se	ppm	1	1000
Tin	Sn	ppm	0.2	500
Strontium	Sr	ppm	0.2	10000
Tantalum	Ta	ppm	0.05	100
Tellurium	Te	ppm	0.05	500
Thorium	Th	ppm	0.01	10000
Titanium	Ti	%	0.005	10
Thallium	Tl	ppm	0.02	10000
Uranium	U	ppm	0.1	10000
Vanadium	V	ppm	1	10000
Tungsten	W	ppm	0.1	10000
Yttrium	Y	ppm	0.1	500
Zinc	Zn	ppm	2	10000
Zirconium	Zr	ppm	0.5	500

NOTE: Four acid digestions are able to dissolve most minerals. However, depending on the sample matrix, not all elements are quantitatively extracted. For example:

- This digestion may not be complete for minerals such as corundum (Al_2O_3), kyanite (Al_2SiO_5) and more complex silicates such as garnet, staurolite, topaz and tourmaline.*
- Potassium may bias low due to the formation of the insoluble perchlorate, which may not be completely decomposed during the leaching process.*
- Low recoveries of Al and Ca may occur if their insoluble fluorides are not completely decomposed during the leaching process.*
- Scandium may not be fully solubilized and may show lower recovery by this digestion. Sc-ICP06 (Lithium Metaborate Fusion, ICP-AES Finish), a method developed for Scandium, can be used as an alternative for this analyte.*
- Four acid digestions can also volatilize certain exploration pathfinder elements, in particular mercury. Mercury is better analyzed by an aqua regia digestion and can be added as a package to this analysis (Package: ME-MS61m).*

APPENDIX F

SAMPLE MAPS WITH SAMPLE ELEMENTAL PPM VALUES

